

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

(Permanent Affiliation to JNTUK, Kakinada), PEDATADEPALLI, TADEPALLIGUDEM-534 101



Department of Computer Science and Engineering

&

Department of Computer Science and Technology

I-SEMESTER

S.No.	Code	Name of the Course	L	T	P	C
1	V20MAT01	Linear Algebra and Differential Equations	3	-	-	3
2	V20CHT01	Engineering Chemistry	3	-	-	3
3	V20ENT01	English for Professional Enhancement	3	-	-	3
4	V20MEL02	Engineering Workshop	1	-	4	3
5	V20CST01	Programming in 'C' for problem Solving	3	-	-	3
6	V20ENL01	Hone Your Communications Skills Lab-I	-	-	3	1.5
7	V20CHL01	Engineering Chemistry Lab	-	-	3	1.5
8	V20CSL01	Programming Lab in 'C' for problem Solving	-	-	3	1.5

Total Contact Hours: 26

Total Credits:19.5

II-SEMESTER

S.No.	Code	Name of the Course	L	T	P	C
1	V20MAT02	Numerical Methods and Vector Calculus	3	-	-	3
2	V20PHT01	Engineering Physics	3	-	-	3
3	V20ECT01	Switching Theory and Logic Design	3	-	-	3
4	V20CST02	Python Programming	3	-	-	3
5	V20MEL01	Engineering Graphics	1	-	4	3
6	V20PHL01	Engineering Physics Lab	-	-	3	1.5
7	V20CSL02	Python Programming Lab	-	-	3	1.5
8	V20ENL02	Hone Your Communications Skills Lab-II	-	-	3	1.5
9	V20CHT02	Environmental Studies	2	-	-	0

Total Contact Hours: 28

Total Credits:19.5

SEMESTER-III (SECOND YEAR)

S.No.	Code	Name of the Course	L	T	P	C
1	V20MBT51	Managerial Economics and Financial Analysis	3	-	-	3
2	V20MAT07	Mathematical Foundation Of Computer Science	3	-	-	3
3	V20CST03	OOPs Through C++	3	-	-	3
4	V20CST04	Data Structures	3	-	-	3
5	V20CST05	Computer Organization and Architecture	3	-	-	3
6	V20CSL03	OOPs Through C++ Lab	-	-	3	1.5
7	V20CSL04	Data Structures Lab	-	-	3	1.5
8	V20CSL05	Linux Shell Scripting Lab	-	-	3	1.5
9		Skill Oriented Course - I	1	0	2	2
10	V20ENT02	Professional Communication Skills -I	2	-	-	0
Total:			18	0	11	21.5

Total Contact Hours: 29

Total Credits:21.5

SEMESTER - IV (SECOND YEAR)

S.No.	Code	Name of the Course	L	T	P	C
1	V20CST06	Design and Analysis of Algorithms	3	-	-	3
2	V20CST07	Software Engineering	3	-	-	3
3	V20CST08	Database Management Systems	3	-	-	3
4	V20CST09	Java Programming	3	-	-	3
5	V20MAT04	Probability and Statistics	3	-	-	3
6	V20CSL06	Statistical Visualization using R Lab	-	-	3	1.5
7	V20CSL07	Database Management Systems Lab	-	-	3	1.5
8	V20CSL08	Java Programming Lab	-	-	3	1.5
9		Skill Oriented Course - II	1	-	2	2
10	V20ENT03	Professional Communication Skills -II	2	-	-	0
Total:			18	0	11	21.5
Student have to do Mini Project / Internship (2 Months) during summer						

Total Contact Hours:29

Total Credits:21.5

Semester	I Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20MAT01
Name of the Course	Linear Algebra and Differential Equations					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Apply matrix technique to solve system of linear equations **(K3)**
CO2: Find Eigen values and Eigen vectors **(K3)**
CO3: Solve the ordinary differential equations of first order & first degree **(K3)**
CO4: Solve the linear differential equations of higher order with constant coefficients. **(K3)**
CO5: Find maxima and minima of functions of two variables. **(K3)**

UNIT-I: System of linear equations: Rank-Echelon form-Normal form – Solution of linear systems – Gauss elimination – Gauss Jordan- Gauss Jacobi and Gauss Seidal methods.

UNIT-II: Eigen values, Eigen vectors and Cayley-Hamilton theorem: Eigenvalues - Eigen vectors– Properties – Cayley-Hamilton theorem (without proof) - Inverse and powers of a matrix by using Cayley-Hamilton theorem.

UNIT-III: Differential equations of first order and first degree: Linear- Bernoulli - Exact-Reducible to exact differential equations -Newton's Law of cooling-Law of natural growth and decay-Orthogonal Trajectories.

UNIT- IV: Linear differential equations of higher order:

Linear non homogeneous differential equations of higher order with constant coefficients involving RHS term of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax}V(x)$, $xV(x)$ - method of variation of parameters.

UNIT- V: Partial differentiation: Introduction to partial differentiation-Total derivative-Functional dependence- Jacobian.-maxima and minima of functions of two variables (without constraints) and Lagrange's method (with constraints).

Text Books

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali, Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India
2. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press
3. Srimanta Pal, Subodh C. Bhunia, Engineering Mathematics, Oxford University Press.
4. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, Delhi.

Semester	I Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CHT01
Name of the Course	Engineering Chemistry					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Solve boiler troubles originated due to poor water quality and suggest suitable water treatment methods.. (K3)
- CO2:** Choose plastics and rubbers for engineering applications.(K3) (K3)
- CO3:** Associate concepts of Electro Chemistry in designing electrochemical energy systems. (K2)
- CO4:** Assess the quality of fuels. (K3)
- CO5:** Apply corrosion principles for protection of metallic structures. (K3)

UNIT-I: WATER TECHNOLOGY: Sources of water; Impurities in water, Hardness of water, Types of Hardness, Units of hardness, Determination of hardness of water, Numerical problems on temporary and permanent hardness.

Boiler troubles: Priming and Foaming, Sludge and Scale formation, Boiler corrosion, Caustic embrittlement. Softening of hard water- Zeolite process and Ion exchange process; Water for drinking purpose, BSI standards of drinking water, Disinfection: Chlorination, Breakpoint chlorination. Desalination - Reverse Osmosis and Electro dialysis.

UNIT-II: POLYMER TECHNOLOGY : Introduction, Polymerization, Mechanism of Free radical addition polymerization; Plastics as engineering materials; Advantages and limitations, Thermoplastics and Thermosetting plastics, Fabrication of plastics (Compression, Injection, Transfer, and Extrusion Moulding) -Preparation, properties and applications of Polythene (HDPE and LDPE), PVC, Bakelite. Elastomers: Disadvantages of natural rubber, Vulcanization of rubber, Advantages of vulcanized rubber, Preparation, properties and applications of Buna -S and Buna-N.

UNIT-III: ELECTRO CHEMISTRY : Galvanic cell, Electrode potential and EMF-Reference electrodes (Calomel and Glass electrodes), Determination of pH of a solution using glass electrode, Conductometric titration (Strong Acid – Strong Base).

Batteries: Types, Primary battery - Li-MnO₂ battery, Secondary batteries - Lead acid battery, Lithium ion battery. Fuel cells: Definition, H₂ - O₂ fuel cell

UNIT-IV: FUEL TECHNOLOGY: Fuels – Characteristics of a good fuel, Classification of fuels, Calorific value (HCV and LCV),Dulong's formula, Numerical problems on HCV and LCV.

Solid fuels: Coal – Proximate and ultimate analysis, Significance of the analyses.

Liquid fuels: Petroleum composition, Classification, Synthetic petrol (Fischer Tropsch and Bergius process), Knocking, Anti knocking agents, Octane and Cetane ratings.

Gaseous fuels – Natural gas, LPG and CNG Biofuels - Biogas, Biodiesel.

UNIT- V : Corrosion: Definition, Theories of Corrosion (Chemical & Electrochemical), Pilling- Bedworth Rule, Galvanic corrosion and Pitting corrosion, Factors which influence the rate of corrosion. Protection from corrosion - Design & Selection of metals, Cathodic protection, Protective coatings – Metallic coatings (Anodic and cathodic coatings), Methods of application of coatings on metals (Galvanizing & Tinning). Nano materials: Introduction, Carbon nanotubes - preparation (Arc discharge, Laser ablation and CVD Method) - Properties and applications of carbon nanotubes.

Text Books

1. Engineering Chemistry by Jain and Jain; Dhanpat Rai Publications & Co.
2. A Text book of Engineering Chemistry by S. S. Dara; S. Chand & Co Ltd.

Reference Books:

1. Engineering Chemistry by Vajiram and others. Wiley India Pvt.Ltd.,
2. Engineering Chemistry by Prasanth Rath, Cengage Learning.
3. Engineering Chemistry by Shikha Agarwal; Cambridge University Press.
4. Engineering Chemistry, by B. Sivasankar, McGraw-Hill.

Semester	I Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20ENT01
Name of the Course	English for Professional Enhancement					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Identify the central theme of the text, use cohesive items for coherence in a paragraph, recognize nouns and basic sentence structures. (K2)
- CO2:** Restate the central idea of the letter by using appropriate vocabulary. Gain mastery over articles and prepositions. (K2)
- CO3:** Find the success formula after reading the text in detail to answer questions. Use appropriate tense and concord, find suitable vocabulary and format to draft letters and e-mails. (K3)
- CO4:** Employ reading skills to comprehend the given biography. Interpret visual information .Use quantifiers appropriately and get acquainted with formal drafting (K3)
- CO5:** Appraise the delivered lecture and text, recognize the contextual vocabulary and prepare poster presentations. (K4)

UNIT-I : A DRAWER FULL OF HAPPINESS (From Infotech English, Maruthi Publications) Vocabulary:

GRE Vocabulary , Antonyms and Synonyms, Word Applications, Verbal Reasoning and Sequencing of Words. Grammar: Nouns: Types of Nouns: Proper Noun, common noun, collective noun, material noun, abstract noun Listening: Listening to short audio texts and identifying the topic, context and specific pieces of information to answer a series of questions both in speaking and writing. Speaking: Self- Introduction and Introducing others. Asking and answering general questions on topics such as home, family, work, studies and interests.

Reading: Skimming text to get the main idea. Scanning to look for specific pieces of information.

Writing: Paragraph Writing

Non- Detailed : The Post Office by Rabindranath Tagore (Macmillan India)

UNIT-II: NEHRU'S LETTER TO HIS DAUGHTER INDIRA ON HER BIRTHDAY

(From Infotech English, Maruthi Publications).

Vocabulary: GRE Vocabulary, Antonyms and Synonyms

Grammar: Articles, Prepositions

Listening: Answering a series of questions about main idea and supporting ideas after listening to audio texts both in speaking and writing.

Speaking: Discussion in pairs/ small groups on specific topics. Functional English: Greeting and Leave Taking.

Reading: Identifying sequence of ideas; Recognizing verbal techniques that help to link the ideas in a paragraph together.

Writing: Identifying the main ideas, Rephrasing and Summarizing them .

Non- Detailed : The Post Office by Rabindranath Tagore (Macmillan India)

UNIT-III: STEPHEN HAWKING - POSITIVITY 'BENCHMARK' (From Infotech English, Maruthi Publications).

Vocabulary: GRE Vocabulary

Grammar: Verbs, Tenses, Concord: Subject - Verb Agreement.

Listening: Listening for global comprehension and summarizing what is listened to both in speaking and writing Speaking: Discussing specific topics in pairs or small groups and reporting what is discussed. Functional English: Complaining and Apologizing.

Reading: Reading a text in detail by making basic inferences –recognizing, and interpreting specific context clues; strategies to use text clues for comprehension, critical reading.

Writing: Letter writing- types, format and principles of letter writing, E-mail Etiquette

Non- Detailed : The Post Office by Rabindranath Tagore (Macmillan India)

UNIT-IV: LIKE A TREE, UNBOWED : WANGARI MAATHAI - BIOGRAPHY

(From Infotech English, Maruthi Publications).

Vocabulary: GRE Vocabulary, Antonyms and Synonyms,

Grammar: Active & Passive Voice

Listening: Making predictions while listening to conversations/ transactional dialogues without video (only audio), listening to audio-visual texts.

Speaking: Role plays for practice of conversational English in academic contexts (formal and informal) - Asking for and Giving Information/Directions. Functional English: Asking for Permissions, Requesting, Inviting.

Reading: Studying the use of graphic elements in text to convey information.

Writing: Data Interpretation – Tree Diagram, Pie chart

Non- Detailed : The Post Office by Rabindranath Tagore (Macmillan India)

UNIT-V: STAY HUNGRY, STAY FOOLISH (From Infotech English, Maruthi Publications).

Vocabulary: GRE Vocabulary, Antonyms and Synonyms

Grammar: Identifying and Correcting Common Errors in Grammar and Usage (articles, prepositions, tenses, subject-verb agreement), Reported Speech.

Listening: Identifying key Terms, Understanding Concepts and Interpreting the Concepts both in speaking and writing.

Speaking: Formal oral presentations on topics from academic contexts. Functional English: Suggesting/Opinion giving.

Writing: Poster Presentation.

Non- Detailed : The Post Office by Rabin dranath Tagore (Macmillan India)

Books Prescribed

“Infotech English”, Maruthi Publications. (Detailed)

“The post Office” by Rabindranath Tagore, Macmillan India(Non -Detailed)

Reference books:

1. Bailey, Stephen. Academic writing: A handbook for international students. Routledge,2014.
2. Chase, Becky Tarver. Pathways: Listening, Speaking and Critical Thinking. Heinley ELT; 2nd Edition, 2018.
3. Skillful Level 2 Reading and Writing Student’s Book Pack (B1) Macmillan Educational.
4. The Official Cambridge Guide to IELTS, for Academic and General Training.(2015)
5. Practical English Usage, Michael Swan, OUP ,1995.

******Note:** The Lessons and GRE Vocabulary has been taken from Infotech English by Maruthi Publications
Non-detailed Text : Post Office by Rabindranath Tagore (Rupa Publications).

Semester	I Sem	L	T	P	C	COURSE CODE
Regulation	V20	1	-	4	3	V20MEL02
Name of the Course	Engineering Workshop					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Prepare different models in the carpentry trade and understand basic concepts of carpentry. **(K3)**

CO2: Develop various basic prototypes in the trade of Tin smithy and understand basic concepts of Tin smithy. **(K3)**

CO3: Prepare various basic prototypes in the trade of fitting and understand basic concepts of fitting. **(K3)**

CO4: Prepare different models in the Black smithy and understand basic concepts of Black smithy. **(K3)**

CO5: Develop various basic House Wiring techniques, Electrical wiring circuits. **(K3)**

CO6: Develop various basic prototype models in Welding and Foundry shop. **(K3)**

Module-I

General safety Considerations during operation of:

Bench Tools, Hammers, Screw Drivers, Punches, Chisels, Scrapers, Scribes, Files, Pliers and Cutters, Wrenches, Hacksaw, Bench Vise, Hand drill, Taps and Dies, Hand Shears, Rules, Tapes and Squares, Soldering Iron, Rivets.

Hand Working Operations:

Sawing, Filing, Threading, Scribing, Shearing, Soldering, Sharpening of hand tools.

Measuring and Gauging:

Calipers, depth Gauge, Feeler Gauge, Micrometers, Vernier Calipers, Vernier Height Gauge, Snap Gauge, Hole Gauge, Bevel Protractor, Dial Indicator, Gauge Blocks and Surface Plate

Module-II

Carpentry:

Introduction, Carpentry Tools, Marking and Layout, Operations.

Sheet Metal Works :

Introduction, Sheet Metal Tools, Marking and Layout, Operations – Bending, Cutting, Rolling.

Fitting :

Introduction, Fitting Tools, Marking and Layout, Operations.

Forging :

Introduction , Forging Tools ,Operations – Upsetting, Drawing, Cutting, Bending, Punching ,Forging Presses and Hammers.

House wiring:

Introduction, House wiring Tools and accessories, Connections, Circuit diagrams.

Metal Joining:

Safety Considerations, Introduction, Soldering, Brazing, Welding – Gas Welding, Arc Welding,

Foundry:

Introduction, Pattern Making, Foundry Tools, Core Making, Melting Furnace – Cupola, Sand Casting Process.

Module-III

Note: At least two exercises to be done from each.

Carpentry

1. T-Lap Joint
2. Cross Lap Joint
3. Dovetail Joint
4. Mortise and Tenon Joint

Tin Smithy

1. Taper Tray
2. Square Box without lid
3. Open Scoop
4. Funnel

Fitting shop

1. V- Fit
2. Square Fit
3. Half Round Fit
4. Dovetail Fit

Black smithy

1. Round rod to Square
2. S-Hook
3. Round Rod to Flat Ring
4. Round Rod to Square headed bolt

House wiring

1. Parallel / Series Connection of three bulbs
2. Stair Case wiring
3. Florescent Lamp Fitting
4. Measurement of Earth Resistance

Welding shop (Arc welding)

1. Butt Joint
2. Lap Joint

Foundry Practice

Preparation of sand mould using split piece pattern and cast the component.

Text Books:

1. A Course in Work shop Technology, Vol.1, Raghuwanshi,Dhanpat Rai &Co.
2. Elements of Workshop Technology, Vol.1, S.K.Hajra Choudary, Asia Publishing House.
3. Production Technology, Vol.1, R.K.Jain and S.C Gupta, Khanna Publications.
4. Workshop Practice Manual,K.Venkata Reddy, B.S.Publications.
5. Workshop Manual, P.Kannaiah, KL.Narayana, Scitech Publications.

Semester	I SEM	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CST01
Name of the Course	Programming in 'C' for problem Solving					
Branch	Common to All					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- | | | |
|-------------|--|------|
| CO1: | Describe various problem solving strategies such as Algorithms and Flowcharts. | (K2) |
| CO2: | Develop various programming constructs using Control Structures. | (K3) |
| CO3: | Construct Programs using modular programming approach. | (K3) |
| CO4: | Illustrate the usage of Arrays, String and pointers. | (K3) |
| CO5: | Construct Programs using Structures, Unions and Files. | (K3) |

UNIT-I: : Problem solving concepts: Algorithms, Flow-charts, Types of Programming Languages, Compiler, Assembler and Linker, Testing and Debugging a program. Introduction to C Programming: Overview and importance of C, C Program Structure, Creation and Compilation of C Programs, Identifiers, Variables, Data types, Constants, Declarations, Input and output statements: Input and output functions...

UNIT-II: Operators: Arithmetic, relational and logical operators, increment and decrement operators, conditional operator, assignment operator, bitwise operators, special operators, expressions, Precedence, Associativity, Order of evaluation, Type conversion, Programming Examples. Control Structures: Conditional statements - If-else, Switch-case constructs, Loops - while, do-while, for..

UNIT-III: Functions: Top down approach of problem solving, standard library functions, user defined functions, parameter passing - call by value, call by reference, return statement, passing arrays as parameters to functions, recursion. Storage Classes: Scope and extent, Storage Classes - auto, extern, static and register.

Understanding pointers: Accessing the address of a variable, declaring pointer variables, initialization of pointer variables, accessing a variable through its pointer, pointer arithmetic.

UNIT-IV: Arrays: Single-Dimensional Arrays, multi-Dimensional Arrays, initialization and accessing individual elements. Strings in C-Concepts, string handling functions. Pointer and arrays, pointers and character strings, array of pointers. Dynamic Memory Allocation: calloc(), malloc() and free()

UNIT-V: Structures: Defining, declaring, initialization, accessing, comparing, operations on individual members, array of structures, structures with in structures, structures and functions, bit fields, Programming Examples. **Unions:** Definition – difference between structures and unions – declaring and accessing unions.– pointers and structures – self-referential structures.

File Processing: Creating and Opening a file, file opening modes, closing a file, input/output operations on files, error handling during I/O operations, random access to files, Command line arguments. Programming Examples.

Text Books:

1. Programming in ANSI C by E Balagursamy, McGraw Hill, 8th Edition.

REFERENCE BOOKS:

1. Let Us C, Yashavant Kanetkar, BPB Publications, 15th Edition
2. Programming in C, Reema Thareja, Oxford.
3. Programming with C, Second edition, Byron S Gottfried, Tata Mc Grawhill
4. Problem Solving and Programm design in C, Hanly J R & Koffman E.B, Pearson Education, 2009.
5. Programming in C, Pradip Dey, Manas Ghosh, Oxford University Press, 2007.
6. Problem Solving Using C: Structured Programming Techniques, Yuksel Uckan.
7. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE.
8. Computer Programming in C – Kerninghan & Ritchie, PHI
9. C: The Complete Reference: Herbert Schildt, Osborne/Mcgraw Hill, Inc.

Semester	I SEM	L	T	P	C	COURSE CODE
Regulation	V20	0	0	3	1.5	V20ENL01
Name of the Course	Hone your Communication Skills, Lab-I					
Branch	Common to All					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Identify suitable expressions to greet people, say good bye to them, introduce one another, listen to consonants. **(K2)**

CO2: Select suitable words to invite someone, accept or decline invitations, listen to..., identify and produce vowel sounds. **(K2)**

CO3: Choose suitable expressions to seek/refuse permissions, to apologize and listen to word accent. **(K3)**

CO4: Find apt expressions to give suggestions, express opinions, use appropriate words to give commands and requests. **(K3)**

CO5: Practise listening to dialogues, role-plays using common vocabulary used in dialogues. **(K3)**

Unit I: Hello, I'm

- Greeting people
- Saying goodbye to people
- Introducing yourself to someone/someone to someone else
- Listening and Identifying Consonants

Unit II: I Would Love to.... but,

- Inviting someone
- Accepting or declining invitations
- Complaining about something
- Listening to, Identifying and Producing Vowel Sounds

Unit-3 With Your Permission I would like to.....

- Seeking Permission
- Granting/refusing permissions
- Apologising
- Listening to syllables and Word Accent and practise.

Unit-4 Why don't we....?

- Making Suggestions
- Agreeing/disagreeing with a suggestion
- Expressing Opinions
- Giving Commands/instructions
- Requesting someone for something

Unit-5 Dialogues

- The norms of dialogues
- Common vocabulary used in dialogues
- Carrying on a dialogue
- Listening to dialogue.

Book Prescribed:

- **Strengthen Your Steps - A multimodal course in communication skills (Maruthi Publications)**

Books for Further Reference

1. Better English Pronunciation (J.D.O'Connor), Cambridge University.
2. English Conversation Practice (A Practical Guide to improve Conversational Skills), Sterling Publishers.
3. Exercise in spoken English, Parts-I-III.CIFEL, Hyderabad, Oxford University Press.

Semester	I SEM	L	T	P	C	COURSE CODE
Regulation	V20	0	0	3	1.5	V20CHL01
Name of the Course	Engineering Chemistry Lab					
Branch	Common to All					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Analyse quantitatively a variety of samples using volumetric methods and instrumental methods.

CO2: Apply volumetric and instrumental methods for the determination of water quality parameters namely Alkalinity, Hardness and pH.

CO3: Prepare polymeric materials and analyse the given coal samples.

LIST OF EXPERIMENTS

1. Introduction to chemistry laboratory – Molarity, Normality, Primary, Secondary standard solutions, Volumetric titrations, Quantitative analysis, Qualitative analysis etc.,
2. Estimation of HCl using standard Na₂CO₃ solution.
3. Estimation of KMnO₄ using standard oxalic acid solution.
4. Determination of alkalinity of a sample of water.
5. Determination of total hardness of water using standard EDTA solution.
6. Estimation of ferrous iron using standard K₂Cr₂O₇ solution.
7. Estimation of copper using standard EDTA solution.
8. Estimation of % available chlorine in bleaching powder.
9. Estimation of pH of the given sample solution using pH meter.
10. Conductometric titration between strong acid and strong base.
11. Proximate analysis of coal.
12. Preparation of phenol – formaldehyde resin.

Text Book:

1. Lab manual prepared by Department of Chemistry, Sri Vasavi Engineering College.

Reference Books:

1. Practical Engineering Chemistry by K. Mulkanti, B.S. Publications.
2. Vogel's Quantitative Chemical Analysis – 5th Edition, Longman.
3. A Text Book on experiments and Calculations Engineering by S.S.Dara, S.Chand & Co Ltd.

Semester	I Sem	L	T	P	C	COURSE CODE
Regulation	V20	-	-	3	1.5	V20CSL01
Name of the Course	Programming Lab in 'C' for problem Solving					
Branch	Common to All					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate problem solving techniques. (K3)
CO2: Construct Programs using the concepts of Arrays, Strings and Pointers (K3)
CO3: Apply the concepts of Functions, Structures and Unions. (K3)
CO4: Use various file processing operations to develop real-time applications (K4)

LIST OF EXPERIMENTS

Tutorial 1: Problem solving using computers.

Lab1: Familiarization with programming environment.

Tutorial 2: Variable types and type conversions.

Lab 2: Simple computational problems using arithmetic expressions.

Tutorial 3: Branching and logical expressions.

Lab 3: Problems involving if-then-else structures switch – case.

Tutorial 4: Loops, while and for loops.

Lab 4: Iterative problems e.g. sum of series.

Tutorial 5: Functions call by value, call by reference

Lab 5: Simple functions.

Tutorial 6: Recursion, structure of recursive calls.

Lab 6: Recursive functions.

Tutorial 7: Pointers.

Lab 7: Programming with pointers.

Tutorial 8: 1D Arrays: searching, sorting.

Lab 8: 1D Array manipulation.

Tutorial 9: 2D arrays.

Lab 9: Matrix problems.

Tutorial 10: String handling.

Lab 10: String handling functions.

Tutorial 11: Structures, unions and dynamic memory allocation.

Lab 11: Structures & unions.

Tutorial 12: File handling, command line arguments.

Lab 12: File operations.

Text Books:

1. Programming in Ansi C by E Balagursamy, McGraw Hill, Eight Edition.

Reference Books:

1. Mastering C, K.R. Venugopal and S.R. Prasad, TMHPublishers.
2. Computer Programming in C, V. Rajaraman, PHI.
3. Programming in C, Stephen G. Kochan, Fourth Edition, PearsonEducation.
4. C- The Complete Reference, Herbert Schildt, Osborne/Mcgraw Hill, Inc.
5. Programming with C, Byron S Gottfried, Second edition, Tata McGrawhill.
6. Programming in C, ReemaThareja, Oxford.
7. Problem Solving and Program design in C, Hanly J R &Koffman E.B, Pearson Education, 2009
8. Programming and Problem Solving Using C, ISRD Group, Tata McGrawHill, 2008.

Semester	II Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	-	-	3	V20MAT02
Name of the Course	Numerical Methods and Vector Calculus					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Compute approximate roots of algebraic and transcendental equations and interpolating polynomial for the given data **[K3]**
- CO2:** Solve ordinary differential equations with initial conditions using numerical methods **[K3]**
- CO3:** Find multiple integrals and improper integrals **[K3]**
- CO4:** Calculate gradient of a scalar function, divergence and curl of a vector function **[K3]**
- CO5:** Apply the knowledge of vector integral concepts to find characteristics of vector fields **[K3]**

UNIT-I: Solution of Algebraic and Transcendental Equations and Interpolation: Introduction- Bisection method – Method of false position– Newton-Raphson method (One variable) - finite differences- forward differences, backward differences – simple relations on forward, backward, central, average and shifting operators - Newton’s formulae for interpolation - Lagrange’s interpolation formula.

UNIT-II: Numerical Integration and solution of Ordinary Differential equations: Trapezoidal rule- Simpson’s $1/3^{\text{rd}}$ and $3/8^{\text{th}}$ rule-Solution of ordinary differential equations by Taylor’s series-Picard’s Method- Euler’s method-Euler’s modified Method – Runge - Kutta method (fourth order).

UNIT-III: Multiple Integrals: Definition of Improper integrals - Double and triple integrals – Change of variables – Change of order of integration.

UNIT-IV: Vector Differentiation: Vector differential operator-Gradient-Divergence-Curl-Laplacian and second order operators -Vector identities.

UNIT- V: Vector Integration: Line integral: Work done – Potential function – Surface and volume integrals - Vector integral theorems: Greens, Stokes and Gauss Divergence theorems (without proof) and related problems.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, **43rd** Edition, Khanna Publishers.
2. N.P.Bali, Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, **10th** Edition, Wiley-
2. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press
3. V.Ravindranath and P.Vijayalakshmi, Mathematical Methods, Himalaya Publishing House, India.
4. Srimanta Pal, Subodh C. Bhunia, Engineering Mathematics, Oxford University Press.
5. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt.Ltd, Delhi.

Semester	II Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20PHT01
Name of the Course	Engineering Physics					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Associate the basic principles of structure of materials, crystallography and X-ray diffraction. **(K2)**
CO2: Prepare the students to the basic concepts of Lasers and their applications in optical fiber communication link **(K3)**
CO3: Indicate the applications of sound waves in various fields **(K2)**
CO4: Interpret wave and particle behavior of matter and relate it to electron theory of metals **(K3)**
CO5: Examine the advanced concepts of engineering materials like Semiconductors, Superconductors and Dielectrics **(K3)**

UNIT-I : CRYSTALLOGRAPHY : Introduction – Space lattice – Basis – Unit Cell – Lattice parameters – Crystal systems- Bravais lattices– Structures and packing fractions of SC,BCC and FCC
RAY DIFFRACTION: Directions and planes in crystals – Miller indices – Separation between successive [h k l] planes – Bragg’s law-Bragg’s x-ray spectrometer.

UNIT-II: LASERS: Introduction –Characteristics of lasers – Spontaneous and Stimulated emission of radiation – Einstein’s coefficients –Pumping schemes– Population inversion– Ruby laser- Helium Neon laser-Applications of LASER.

FIBEROPTICS: Introduction–Structure of an optical fiber-Principle of Optical Fiber – Acceptance angle and acceptance cone – Numerical aperture- Basic optical communication system-Advantages of optical fibers over conventional transmission lines.

UNIT–III: ACOUSTICS: Introduction - Sound absorption- Absorption coefficient- Reverberation- Reverberation Time – Basic requirements for constructing an acoustically good hall - Sabine’s formula- Factors affecting acoustics of buildings and their remedial measures.

ULTRASONICS: Introduction- Production of Ultrasonic Waves Using Piezoelectric Effect and Magnetostriction Method- Non-Destructive Testing - Pulse Echo Technique – Applications of ultrasonics.

UNIT – IV: QUANTUM MECHANICS: Introduction-de-Broglie’s concept of matter waves – Schrodinger’s Time Independent & time dependent wave equations –Physical significance of the wave function- Particle in a one dimensional potential box.

FREE ELECTRON THEORY: Classical free electron theory (qualitative) – Assumptions and failures-Quantum free electron theory – Equation for electrical conductivity based on quantum free electron theory- Density of states (3D) - Fermi energy- Fermi – Dirac distribution.

BAND THEORY OF SOLIDS: Bloch’s function (qualitative) – Kronig – Penney model (qualitative)–formation of energy bands in crystalline solids based on Kronig Penny model – E vs K diagram- v vs K diagram- effective mass of an electron- Classification of crystalline solids-concept of hole.

UNIT – V : SEMICONDUCTOR PHYSICS: Introduction - Types of Semiconductors- Intrinsic Semiconductors- Carrier concentration– Expression for Conductivity- Extrinsic semiconductors-Carrier concentrations- Dependence of Fermi energy on carrier concentration and temperature- Drift and diffusion currents- Einstein's Equation- Hall Effect- Hall coefficient- Applications of Hall Effect. **SUPERCONDUCTIVITY:** Introduction- General properties – Meissner effect - Type I and Type II Superconductors- BCS Theory – Josephson effects (AC and DC) - Applications of superconductors.

DIELECTRIC PROPERTIES: Introduction- Types of polarizations- Electronic, Ionic and Orientation polarizations (qualitative) – Internal electric field – Clausius- Mossotti Equation.

Text Books:

1. A Text book of Engineering Physics, M.N. Avadhanulu and P.G.Kshirasagar, S.Chand Publications.
2. Engineering Physics DK Bhattacharya, Poonam and Tandon Publications.

Reference books:

1. Solid state Physics, A.J. Dekker by McMillan India Ltd.
2. Introduction to Solid state Physics, Charles Kittel, Wiley India Pvt. Ltd.
3. Solid state Physics, S.O. Pillai by New Academic Science.
4. Basic Engineering Physics, Dr.P. Sreenivasa Rao, Himalaya Publishers.
5. Engineering Physics, V. Rajendran, Mc GrawHill.
6. Engineering Physics, Sanjay D Jain and Girish G Sahasrabudhe., University Press.
7. Engineering Physics, Gaur and Gupta, Dhanpat Rai Publications.
8. Engineering Physics, P.K. Palanisamy, Sci Tech Publishers.

Semester	II Sem	L	T	P	C	COURSE CODE
Regulation	V20	-	-	3	1.5	V20ECT01
Name of the Course	Switching Theory and Logic Design					
Branch	Common to EEE, ECE, ECT, CSE & CST					

Syllabus Details

Course Outcomes: At the end of the Course student will be able to:

- CO1:** Explain the different types of number Systems, number conversions, codes and logic Gates. **(K2)**
CO2: Apply the concepts of Boolean algebra and use the knowledge of K-maps and tabular method for minimization of Boolean expressions.. **(K3)**
CO3: Construct the higher order modules from their lower order structures of various M combinational logic circuits. **(K3)**
CO4: Explain the concept of various flip flops. **(K2)**
CO5: Develop various sequential circuits like registers, counters and various Finite State Machine Models **(K3)**

UNIT– I: Number Systems & Codes: Representation of numbers of different radix, conversation from one radix to another radix, r and (r-1)'s compliment of signed members. Basic logic operations -NOT, OR, AND, Universal building blocks, EX-OR, EX-NOR – Gates.

Binary Codes: BCD, Excess-3, Graycode, 2421, 84-2-1, error detection, error correction codes - Hamming Code

UNIT– II: Minimization Techniques : Boolean theorems, principle of complementation & duality, De-morgan theorems, minimization of logic functions using Boolean theorems, Standard SOP and POS, Forms, NAND-NAND and NOR-NOR realizations, minimization of switching functions using K-Map up to 5 variables, tabular minimization.

UNIT– III: Combinational Logic Circuits Design : Half adder, full adder, half subtractor, full subtractor, Ripple Carry adder and subtractor, 4 bit binary adder-subtractor circuit, BCD adder circuit, Excess 3 adder circuit, Design of decoder, demultiplexer, 7 segment decoder, Implementation of higher order circuits using lower order circuits for MUX, DEMUX, DECODER, realization of Boolean functions using decoders and multiplexers, priority encoder.

UNIT– IV: Sequential Circuits –I : Classification of sequential circuits (synchronous and asynchronous); basic flip-flops, truth tables and excitation tables (Nand RS latch, nor RS latch, RS flip-flop, JK flip-flop, T flip-flop, D flip-flop with reset and clear terminals). Asynchronous Inputs (Preset and Clear), Race around condition, Master Slave JK Flip flop, Conversion from one flip-flop to another flip-flop.

UNIT– V: Sequential Circuits –II : Design of ripple counters, design of synchronous counters, Johnson counter, ring counter. Design of registers-Buffer register, control buffer register, shift register, bi-directional shift register, universal shift register. **Finite State Machine:** Introduction to Mealy and Moore Finite state Machines

Text Books

1. Digital Design by M. Morris Mano, Michael D. Ciletti, PEA.
2. Fundamentals of Logic Design, 5/e Roth, Cengage.
3. Modern Digital Electronics by RP Jain, TMH

Reference Books

1. An Engineering Approach to Digital Design, William I. Fletcher, Pearson edition.
2. Switching Theory and Logic Design by A. Anand Kumar
3. Switching & Finite Automata Theory, 2nd Edition, Zvi Kohavi, TMH, 1978.

Semester	II Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CST02
Name of the Course	Python Programming					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: At the end of the Course student will be able to:

- | | |
|--|-------------|
| CO1: Illustrate basic concepts of Python Programming | (K2) |
| CO2: Describe control structures in python. | (K2) |
| CO3: Construct python programs using structured data types. | (K3) |
| CO4: Demonstrate functions and packages | (K3) |
| CO5: Develop programs on Files, Exception handling and OOPs Concepts. | (K3) |

UNIT-I: Introduction to Python, Data Types & Operators: Basics of python programming: Features of python – History of Python - Python installation and execution - Data types – Identifiers - variables – type conversions- Literals, Constants – Numbers – Strings. I/O statements. Operators and expressions, operator precedence – expression evaluation.

UNIT-II: Control Structures: Decision Control statements: conditional (if), alternative (if-else), chained conditional (if-elif-else); Iteration: while loop, for loop, nested for loop, range function, break, continue and pass statements.

UNIT-III: Structured Data Types: Lists: list operations, list slices, list methods, cloning lists, list parameters. Tuples: tuple assignment, tuple as return value. Set: Set Creation, Set Operations. Dictionaries: Creation, operations; comprehension, operations on strings.

UNIT-IV: Functions& modules: Introduction - Function Declaration &Definition - Function Call – Variable Scope and Lifetime - The return statement-More on Defining Functions - Lambda Functions or Anonymous Functions - Documentation Strings- Modules – Packages.

UNIT-V: Files & Exception Handling: Introduction - Types of files - Text files - reading and writing files; Errors and exceptions handling. OOPS concepts Classes, Methods, Constructor, Inheritance, Overriding Methods, Data hiding, TKINTER.

Text Books:

1. “Python Programming using problem solving Approach” ReemaThareja, Oxford University Press –2017.
2. Python with Machine Learning by “A.Krishna Mohan, Karunakar&T.Murali Mohan” by S. ChandPublisher-2018.

Reference Books:

1. Allen B. Downey, “Think Python: How to Think Like a Computer Scientist“, 2nd edition, Updated for Python 3, Shroff /O‘Reilly Publishers,
2. 2016 (<http://greenteapress.com/wp/think-python>)
3. Guido van Rossum and Fred L. Drake Jr, —An Introduction to Python – Revised and updated for Python 3.2, Network Theory Ltd.,2011.
4. John V Guttag, —Introduction to Computation and Programming Using Python“, Revised and expanded Edition, MIT Press ,2013.

Semester	II Sem	L	T	P	C	COURSE CODE
Regulation	V20	1	-	4	3	V20MEL01
Name of the Course	Engineering Graphics					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Understand the basic commands in CAD Software and draw the conic sections. **(K3)**
CO2: Construct different types of scales and special curves. **(K3)**
CO3: Draw the projections of the points and lines. **(K3)**
CO4: Develop the projections of planes and surfaces of regular solids. **(K3)**
CO5: Draw the Isometric projections and conversion of views **(K3)**

UNIT-I: Introduction to CAD Software: CAD Software Mechanical Desktop, Draw, Modify, Dimension tool bars, Annotations, Layers, ISI conventions in drawing. CONIC SECTIONS – Ellipse, Parabola and Hyperbola

UNIT-II: SPECIAL CURVES & SCALES: Special Curves – cycloid, epicycloids, hypocycloid; Scales – Plain, Diagonal and Vernier Scales.

UNIT-III: ORTHOGRAPHIC PROJECTIONS: Introduction to Orthographic Projections- Projections of Points, Projection of lines inclined to both the planes.

UNIT-IV: PROJECTION OF PLANES :Inclined to both the Planes.

PROJECTION OF REGULAR SOLIDS :Prisms, Pyramids, Cones and Cylinders with the axis inclined to one of the planes, Development of Surfaces of regular solids.

UNIT-V: ISOMETRIC PROJECTIONS: Principles of Isometric projection – Isometric Scale, Isometric Views, Conventions; Isometric Views of lines, Planes, Simple solids. Conversion of Isometric Views to Orthographic Views and Vice-versa.

Text Books:

1. Engineering Drawing by N.D. Bhat, Chariot Publications, 53rdEdition-2014
2. Engineering Drawing by Agarwal & Agarwal, Tata McGraw Hill Publishers, 2ndEdition- 2016

Reference Books:

1. EngineeringDrawingbyK.L.Narayana&P.Kannaiah,ScitechPublishers,2ndEdition-2014
2. Engineering Graphics for Degree by K.C. John, PHIPublishers-2014
3. Engineering Graphics by PI Varghese, McGrawHillPublishers-2013
4. EngineeringDrawingAutoCad–KVenugopal,V.PrabhuRaja,NewAge,5thEdition-2015

Semester	II Sem	L	T	P	C	COURSE CODE
Regulation	V20	-	-	3	1.5	V20PHL01
Name of the Course	Engineering Physics Lab					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Analyze the physical principle involved in the various instruments; also relate the principle to new application. **(K4)**

CO2: Demonstrate the various experiments in the areas of optics, mechanics and Electronics in all branches of engineering. **(K3)**

CO3: Think innovatively and also apply the creative skills that are essential for engineering. **(K4)**

LIST OF EXPERIMENTS

(Any eight of the following to be done)

1. Determination of Rigidity modulus of a material – Torsional Pendulum
2. Determination of acceleration due to gravity – Compound Pendulum
3. Verification of laws of vibrations in stretched strings – Sonometer
4. Determination of velocity of sound – Volume Resonator
5. Verification of Magnetic field Induction along the axis of current carrying coil – Stewart and Gee's apparatus.
6. Determination of Planck's constant using photocell.
7. Determination of wave length of laser source using diffraction grating.
8. Determination of frequency of electrically driven tuning fork - Melde's experiment – Transverse and longitudinal modes.
9. Study of V/I Characteristics of Zener diode
10. Draw the frequency responsive curves of L-C-R Series Resonance Circuit.
11. Determination of Energy band gap of a Semiconductor p-n junction
12. Characteristics of Thermistor – Negative Temperature Coefficient of resistivity.

Virtual labs: (Any two of the following to be done)

1. Crystal Structure.
2. Numerical Aperture of an Optical Fiber.
3. Photo-Electric Effect.
4. Hall Effect.

Semester	II Sem	L	T	P	C	COURSE CODE
Regulation	V20	0	0	3	1.5	V20CSL02
Name of the Course	Python Programming Lab					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate Basic Python Programs. (K3)
CO2: Construct control structures in python (K3)
CO3: Demonstrate functions and packages. (K3)
CO4: Construct python programs using structured data types. (K3)
CO5: Demonstrate Text Files and exception handling. (K3)

LIST OF EXPERIMENTS

Exercise 1 – Basics

- A sample Python Script using command prompt, Python Command Line and IDLE
- A program to purposefully raise an Indentation Error and correct it.

Exercise 2 - Operations

- A program to compute distance between two points taking input from the user (Pythagorean Theorem)
- A program on add.py that takes 2 numbers as command line arguments and prints its sum.

Exercise 3- Control Flow

- A Program to implement for checking whether the given number is a even number or not.
- A program to construct reverse the digits of a given number and add it to the original, If the sum is not a palindrome repeat this procedure.
- A program using a while loop that asks the user for a number, and prints a countdown from that number to zero.

Exercise 4 - Control Flow – Continued

- A program to construct the following pattern, using a nested for loop.

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

- By considering the terms in the Fibonacci sequence whose values do not exceed four million, find the sum of the even-valued terms.

Exercise 5- Structured Data types

- A program to count the number of strings where the string length is 2 or more and the first and last character are same from a given list of strings.
- a program to develop unzip a list of tuples into individual lists and convert them into dictionary.

Exercise 6 -Structured Data types Continued

- a) A program to count the numbers of characters in the string and store them in a dictionary data structure
- b) A program to use split and join methods in the string and trace a birthday with a dictionary data structure.

Exercise 7 – Problem Solving using Functions

- a) Find mean, median, mode for the given set of numbers passed as arguments to a function
- b) Develop a function nearly equal to test whether two strings are nearly equal. Two strings a and b are nearly equal when a can be generated by a single mutation on b.
- c) Develop a Recursive Function to find the Factorial of a given number.
- d) Develop function to compute gcd, lcm of two numbers. Each function shouldn't exceed one line.

Exercise 8– Modules

- a) Install packages requests, flask and explore them using(pip)
- b) A program to implement a script that imports requests and fetch content from the page. Eg. (Wiki)
- c) Develop a simple script that serves a simple HTTP Response and a simple HTML Page

Exercise 9 - Files

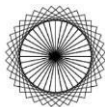
- a) A program to count frequency of characters in a given file. Can you use character frequency to tell whether the given file is a Python program file, C program file or a text file?
- b) A program to compute the number of characters, words and lines in a file.

Exercise 10 -OOP

- a) Class variables and instance variable and illustration of self-variable
 - i) Robot
 - ii) ATM Machine

Exercise - 11 GUI, Graphics

- a) Develop a GUI for an Expression
- b) A program to implement the following figures using turtle



Text Books:

1. “Python Programming using problem solving Approach” Reema Thareja, Oxford University Press – **2017.**
2. Python with Machine Learning by “A.Krishna Mohan, Karunakar &T.Murali Mohan” by S. Chand Publisher-**2018.**

Semester	II Sem	L	T	P	C	COURSE CODE
Regulation	V20	-	-	3	1.5	V20ENL02
Name of the Course	Hone your Communication Skills Lab-II					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Collect suitable expressions and vocabulary to participate in JAM. Identify root words. (K1)
- CO2:** Prepare, face and perform well in interviews with required etiquette. (K1)
- Use appropriate telephone etiquette to succeed in telephonic interviews (K3)
- CO3:** Show team spirit and communicative skills in group discussion. (K3)
- CO4:** Arrange ideas and prepare to give presentations in a professional manner. (K3)
- CO5:** Debate rationally and cogently while putting forth the ideas. (K4)

UNIT-I: JAM Session & Root Words

- Preparation for JAM Session
- Participation in JAM
- Root words from Word power made easy by Norman Lewis

UNIT-II: Interviews

- Guidelines for facing interviews
- Three R's of interviews
- Practice Activity (Mock Interviews)
- Root words from Word power made easy by Norman Lewis
- Telephone Etiquette
- Preparing for telephonic interviews
- Acing interviews
- Practice Activity (Mock Interviews)
- Root words from Word power made easy by Norman Lewis

UNIT-III: Group Discussions

- Tips to participate in Group Discussion
- Practice Activity
- Root words from Word power made easy by Norman Lewis

UNIT-IV: Presentation and Public Speaking

- Three P's of Presentation
- Do's and Don'ts in a Power-point Presentation
- Oral Presentations
- Introduction to Public Speaking
- Strategies for successful Public Speaking
- Practice Activity

UNIT-V: Debate

- Introduction to Debate
- Parts of a Debate
- Guidelines to participate in a Debate
- Practice Activity

Book Prescribed:

Strengthen Your Steps - A multimodal course in communication skills (Maruthi Publications)

Books for further Reference:

1. English Language Communication Skills, Lab Manual cum Workbook (with CD),Cengage Learning.
2. The Students Companion –Wilfred D. Best (New Edition) – Harper,CollinsPublishers,2012.
3. Hewings, Martin. Cambridge Academic English (B2).CUP, 2012. 4. Lewis, Norman. Word Power Made Easy, GoyalSaab,Publications

Semester	II Sem	L	T	P	C	COURSE CODE
Regulation	V20	2	-	-	0	V20CHT02
Name of the Course	Environmental Studies					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Recognise the importance of environment and ecosystem services. **(K2)**
CO2: Identify the characteristic features, uses and impact of overutilization of natural resources **(K2)**
CO3: Explain biodiversity, biodiversity services and conservation of biodiversity **(K2)**
CO4: Report the causes and impacts of various pollutions. **(K2)**
CO5: Illustrate social and global environmental issues; sustainable development practices. **(K2)**

UNIT-I: MULTIDISCIPLINARY NATURE OF ENVIRONMENT & ECOSYSTEM: Definition, Scope and importance of environment, Types of environment, Multidisciplinary nature of Environmental Studies, Components of environment. Ecosystem-Concept of an Ecosystem, Structure and function of an Ecosystem, Food chain & food web, Ecological Pyramids, Structure and function of Forest, Desert, Pond and Marine ecosystem.

UNIT-II: NATURAL RESOURCES: Forest Resources: Uses, Overexploitation, Deforestation. Water resources: Aquifers, Dams and benefits, Conflicts over water. Mineral resources: Uses, Overexploitation, Environmental impact of extraction and use of mineral resources. Land resources: Degradation, Soil erosion and desertification, Landslides. Renewable Energy resources: Solar energy, Geo thermal energy, Tidal Energy.

UNIT-III: BIODIVERSITY AND ITS CONSERVATION: Definition, Levels of Biodiversity, Values of Biodiversity, Hotspots of Biodiversity, Threats to Biodiversity, Endangered and Endemic species of India, In-situ and Ex-situ Conservation.

UNIT-IV: ENVIRONMENTAL POLLUTION : Definition of pollution, Air pollution- Types of Air pollutants, Effects and control measures; Water pollution- Causes, Effects and control measures; Soil pollution; Biomedical waste; Industrial waste- Process of waste management, Sanitary land fill, Incineration, 3R strategy; E- Waste and its management.

UNIT-V: SOCIAL AND GLOBAL ENVIRONMENTAL ISSUES & ACTS : Women Education, Value education, Role of information technology on environment and human health, Acid rains, Global warming, Ozone layer depletion. Population growth. Importance of environmental legislation, Environmental Protection Act, Air Act (Prevention and control of pollution), Water Act.

Text Books:

1. Environmental Studies, Fourth Edition, ANubhaKaushik, C P Kaushik, New Age International Pubishers.
2. A Textbook of Environmental Studies, Shashi Chawla, TMH, NewDelhi.
3. Fundamentals of Environmental Studies, DD Mishra, S Chand & Co.Ltd.
4. Textbook of Environmental Science, DR M. Anjireddy, B.S Publications, Hyderabad.

Semester	III Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20MBT51
Name of the Course	Managerial Economics and Financial Analysis					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Understand the basic concepts of managerial economics, demand, elasticity of demand and methods of demand forecasting. **(K2)**
- CO2:** Interpret production concept, least cost combinations and various costs concepts in decision making. **(K3)**
- CO3:** Differentiate various Markets and Pricing methods along with Business Cycles **(K2)**
- CO4:** Prepare financial statements and its analysis. **(K3)**
- CO5:** Assess various investment project proposals with the help of Capital Budgeting techniques for decision making. **(K3)**

UNIT-I: Introduction to Managerial Economics and demand Analysis: Definition of Managerial Economics and Scope-Managerial Economics and its relation with other subjects-Concept of Demand-Types-Determinants-Law of Demand and its Exceptions-Elasticity of Demand-Types and Measurement- Demand forecasting and its Measuring Methods.

UNIT-II: Production and Cost Analysis: Production function-Iso-quants and Iso-cost-Law of Variable proportions-Cobb-Douglas Production function-Economies of Scale-Cost Concepts- Opportunity Cost-Fixed vs Variable Costs-Explicit Costs vs Implicit Costs- Cost Volume Profit analysis- Determination of Break-Even Point- BEP Chart (Simple Problems).

UNIT-III: Introduction To Markets, Pricing Policies & forms of Organizations and Business Cycles: Market Structures: Perfect Competition, Monopoly, Monopolistic and Oligopoly – Features – Price, Out-put Determination – Methods of Pricing: Evolution of Business Forms - Features of Sole Trader – Partnership – Joint Stock Company – State/Public Enterprises. Business Cycles – Meaning and Features – Phases of Business Cycle.

UNIT-IV: Introduction to Accounting & Financing Analysis: Introduction to Double Entry System – Preparation of Financial Statements- Trading Account, Profit & Loss Account and Balance Sheet - Ratio Analysis – (Simple Problems).

UNIT-V: Capital and Capital Budgeting: Capital Budgeting: Meaning of Capital-Capitalization-Meaning of Capital Budgeting-Need for Capital Budgeting-Techniques of Capital Budgeting-Traditional and Modern Methods.

Text Books

1. Dr. N. AppaRao, Dr. P. Vijay Kumar: 'Managerial Economics and Financial Analysis', Cengage Publications, New Delhi – 2011
2. Dr. A. R. Aryasri – Managerial Economics and Financial Analysis, TMH 2011
3. Prof. J.V.Prabhakararao, Prof. P. Venkatarao. 'Managerial Economics and Financial Analysis', Ravindra Publication.

References:

1. Dr. B. Kuberudu and Dr. T. V. Ramana: Managerial Economics & Financial Analysis, Himalaya Publishing House, 2014.
2. V. Maheswari: Managerial Economics, Sultan Chand, 2014
3. Suma Damodaran: Managerial Economics, Oxford 2011.
4. Vanitha Agarwal: Managerial Economics, Pearson Publications 2011.
5. Sanjay Dhameja: Financial Accounting for Managers, Pearson
6. Maheswari: Financial Accounting, Vikas Publications.
7. S. A. Siddiqui & A. S. Siddiqui: Managerial Economics and Financial Analysis, New Age International Publishers, 2012
8. Ramesh Singh, Indian Economy, 7th Edn., TMH 2015
9. Pankaj Tandon A Text Book of Microeconomic Theory, Sage Publishers, 2015
10. Shailaja Gajjala and Usha Munipalle, Universities press, 201

Semester	III Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20MAT07
Name of the Course	Mathematical Foundation of Computer Science					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate the concepts associated with propositions and mathematical logic. **(K3)**
- CO2:** Demonstrate the basic concepts associated with relations, functions and their applications. **(K3)**
- CO3:** Solve recurrence relations using various methods. **(K3)**
- CO4:** Apply techniques of graphs for real-time problems. **(K3)**
- CO5:** Construct minimal spanning tree by using different algorithms. **(K3)**

UNIT-I : Mathematical Logic: Statements and Notation , Connectives, Well Formed Formulas ,Truth tables, Tautologies, Equivalence of formulas, Tautological Implications, Normal forms, Theory of inference for Statement Calculus, Indirect Method of Proof. Predicate calculus-Predicates, quantifiers, universe of discourse.

UNIT-II: Set Theory and Relations: Operations on Sets, Principle of Inclusion and Exclusion, Relations, Properties of Binary Relations in a set, Transitive Closure, Relation Matrix and Digraph, Equivalence, Partial Ordering Relations, Hasse Diagrams, Lattice and its Properties, Functions, Bijective Functions, Composition of Functions.

UNIT-III: Recurrence relations: Generating Function of Sequences, Calculating Coefficient of generating functions, Recurrence relations, solving recurrence relation by substitution and Generating functions, the method of Characteristic roots, Solution of Inhomogeneous Recurrence Relation.

UNIT-IV: Graph Theory: Basic Concepts of graph, Representing graphs, Sub graphs, Isomorphic graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs, Planar graphs, Graph Coloring, Chromatic Number. (Theorems without proofs)

UNIT-V: Trees: Spanning Trees, minimal Spanning Trees, BFS, DFS, Kruskal's Algorithm, Prim's Algorithm, Binary trees, Planar Graphs.

Text Books:

- Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, 1st Edition, Tata McGraw Hill.
- Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.
- Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel, T.P. Baker, 2nd Edition, Prentice Hall of India.

Reference Books:

- Elements of Discrete Mathematics -A Computer Oriented Approach, C. L. Liu and D. P. Mohapatra, 3rd Edition, Tata McGraw Hill.
- Discrete Mathematics with Combinatorics and Graph Theory, Santha, 1st Edition Cengage Learning.

Semester	III Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CST03
Name of the Course	OOPs Through C++					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Differentiate Procedural Oriented Programming and Object-Oriented Programming. **(K2)**
- CO2:** Develop programs using Classes and Objects. **(K3)**
- CO3:** Demonstrate Constructors, destructors & Operator-Overloading. **(K3)**
- CO4:** Construct Classes using inheritance and Exceptions. **(K3)**
- CO5:** Demonstrate Files and Generic Programming. **(K3)**

UNIT-I: Introduction to Object-Oriented Programming – Programming Paradigms, Data Types, Variables, Constants, Operators, Decision Statements & Control Structures, Arrays, Namespace, Default Arguments, Constant Arguments, Parameter passing techniques, Features of Object-Oriented Programming.

UNIT-II: Introduction to Classes and Objects: Defining Classes & Objects, Access specifiers, Scope Resolution Operator, Static Member variables, Static Member Functions, Array of Objects. Inline Functions, Overloading Member Functions, Objects as Function Arguments, Friend Functions, Friend Class, Local Class, Empty Class, Nested Classes, Return by Reference.

UNIT-III: Introduction to Constructors: Characteristics, Constructor with Default Arguments, Parameterized Constructors, Overloading Constructors, Copy Constructor, Dynamic Constructors and Destructors, Anonymous Objects. Introduction to operator Overloading, Rules for Overloading Operators, Overloading Unary & Binary Operators, this keyword, Constraint on Increment and Decrement Operators, Overloading with Friend Functions, Type Conversions.

UNIT-IV: Inheritance: Base class and Derived class, Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Hybrid Inheritance, Virtual Base Classes, Constructor in Derived Classes. qualifier classes, Significance of Virtual Functions, Early Vs Late Binding, Pure Virtual Functions, Virtual Destructor. **Exception handling:** Principles of Exception Handling, Keywords, Exception Handling Mechanism, Multiple Catch Statements, Catching Multiple Exceptions, Re-throwing Exception.

UNIT-V: Files: File Opening Modes, File Stream Classes, I/O manipulators, Classes for File Handling, Sequential Access Files, Random Access Files, Error Handling Functions.

Generic Programming with Templates: Need for Templates, Class Templates, Function Templates, overloading Template Functions. Introduction to Standard Template Library, Sequential Containers & Associative Containers.

Text Books:

1. Programming in C++, Ashok N Kamthane, 2nd Edition, Pearson.
2. C++ How to Program, Paul J. Deitel, Harvey Deitel, 6th Edition, PHI publication.

Reference Books:

1. Object Oriented Programming C++, Joyce Farrell, Cengage.
2. Mastering C++, Venugopal, Raj Kumar, Ravi Kumar, TMH.
3. The Complete Reference C++, Herbert Schildt, 4th Edition, Mcgraw Hill.
4. Object Oriented Programming With C++, R. Subburaj, Vikas Publishing House.

Semester	III Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CST04
Name of the Course	Data Structures					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Illustrate the time and space complexities for searching and sorting algorithms. **(K2)**
CO2: Demonstrate linked lists and their applications. **(K3)**
CO3: Demonstrate Stacks and Queues. **(K3)**
CO4: Illustrate basic operations on binary trees. **(K3)**
CO5: Demonstrate Graphs and their applications. **(K3)**

UNIT-I: Introduction, searching and sorting: Introduction to Data Structures, Types of Data Structures, Performance Analysis: Space complexity, time complexity, asymptotic notation. **Searching:** Linear, Binary and Fibonacci search. **Sorting:** Bubble sort, Selection sort, Insertion sort, radix sort, quick sort, and merge sort.

UNIT-II: Single linked list: Representation of node, operations on single linked list, **Double linked list:** Representation of node, operations on double linked list. **Circular linked List:** Representation of node and its operations.

UNIT-III: Stacks: Definition, Stack ADT, array representation, linked list representation, Towers of Hanoi, infix to postfix conversion, expression evaluation. **Queues:** definition, Queue ADT, Array representation, linked list representation, operations on queues, Applications of Queues, Circular Queue.

UNIT-IV: Trees: Introduction: Terminology, representation of trees, **Binary trees:** abstract data type, Properties of binary trees, binary tree representation, **Tree Traversals:** Inorder, Preorder, Postorder. **Binary search trees:** Definition, searching BST, insert into BST, delete from a BST, Height of a BST.

UNIT-V: Graph: Introduction, definition, types of Graphs, Graph Representation, operations. **Graph Traversal Techniques:** Breadth First Search, Depth First Search **Spanning Trees:** minimum cost spanning tree, Prim's and Kruskal's algorithms, Single source shortest Path and all pair shortest path algorithms.

Text Books:

1. Data Structures, algorithms and applications in C, SartajSahni, Universities press, Second Edition.
2. Fundamentals of Data Structures in C, Ellis Horowitz, SartajSahni and Dinesh Mehta, 2nd Edition, Universities Press (India) Pvt. Ltd.

Reference Books:

1. An Introduction to Data Structures with Application, Jean-Paul Tremblay , Paul Sorenson, Second Edition.
2. Fundamentals of Data Structures and algorithms by C V Sastry, Rakesh Nayak, Ch. Raja Ramesh, IK Publications, new Delhi.
3. Data structures using C and C++, Langsam, Augenstein and Tanenbaum, PHI.
4. Problem solving with C++, The OOP, Fourth edition, W.Savitch, Pearson education.

Semester	III Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CST05
Name of the Course	Computer Organization and Architecture					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- | | | |
|-------------|--|-------------|
| CO1: | Illustrate Basic structure of Computers, Instruction types and their addressing modes. | (K2) |
| CO2: | Describe the different modes of Input / Output transfer. | (K2) |
| CO3: | Illustrate different types of Memory. | (K2) |
| CO4: | Describe the different types of Control Unit techniques. | (K2) |
| CO5: | Explain the Concepts of Pipelining and Parallel Processing | (K2) |

UNIT-I: Introduction: Functional Units, Basic Operational Concepts, Bus Structures.

Instruction Sequencing and Addressing Modes: Instructions and Instruction Sequencing, Addressing modes, Basic Input/output Operations.

UNIT-II: Input/output Organization: Accessing Input/output devices, Interrupts- Interrupt Hardware, Enabling and Disabling Interrupts, Handling Multiple Devices, Direct Memory Access, Buses- Synchronous and Asynchronous.

UNIT-III: Memory Organization: Memory Hierarchy, Main Memory, Auxiliary Memory, Associative memory, Cache Memory. (Morris Mano)

UNIT-IV: Processing Unit: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control, Microprogrammed Control-Microinstructions, Microprogram Sequencing.

UNIT-V: Pipelining: Basic Concepts, Data Hazards, Instruction Hazards.

Parallelism: Parallel processing challenges – Flynn’s classification – SISD, MIMD, SIMD, SPMD, and Vector Architectures - Hardware multithreading – Multi-core processors and other Shared Memory Multiprocessors - Introduction to Graphics Processing Units, Clusters, Warehouse Scale Computers and other Message-Passing Multiprocessors.

Text Books:

- Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 5th Edition, McGraw Hill Education. Computer System Architecture, M. Morris Mano, 3rd Edition, Pearson Education..
David A. Patterson and John L. Hennessy, Computer Organization and Design: The Hardware/Software Interface, Fifth Edition, Morgan Kaufmann / Elsevier, 2014.

Reference Books:

- Computer Organization and Architecture, William Stallings, 10th Edition, Pearson Education.
- Computer Architecture and Organization, John P. Hayes, 3rd Edition, McGraw Hill Education.

Semester	III Sem	L	T	P	C	COURSE CODE
Regulation	V20	0	0	3	1.5	V20CSL04
Name of the Course	Data Structures Lab					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- | | |
|--|-------------|
| CO1: Construct Programs on Sorting and Searching Techniques. | [K3] |
| CO2: Illustrate Various Operations On Linked Lists. | [K3] |
| CO3: Develop Programs On Stacks, Queues and Their Applications. | [K3] |
| CO4: Develop Various Operations on Trees and Graphs | [K3] |

LIST OF EXPERIMENTS

1. Practice following Sorting Techniques
(A) Selection Sort (B) Quick Sort (C) Merge Sort
2. Practice following Searching Methods
(A) Linear Search (B) Binary Search.
3. Develop program for
Single Linked List and Its Operations. (Create, Insert, Delete, Display)
4. Develop program for Double Linked List and Its Operations.
5. Construct Stack along with their operations using Arrays.
6. Construct Queue along with their operations using Arrays.
7. Develop Circular Queue using Arrays.
8. Construct Queue along with their operations using Single Linked List.
9. Construct Binary Search Tree and Its Operations using double linked list.
10. Demonstrate Depth First Search and Breadth First Search Algorithm.
11. Develop Minimum Spanning Tree using Prim's Algorithm.
12. Develop Minimum Spanning Tree Kruskal's Algorithm.

Text books:

1. Data Structures, algorithms and applications in C++, Sartaj Sahni, Universities press, Second Edition.
2. Fundamentals of Data Structures in C++, Ellis Horowitz, Sartaj Sahni and Dinesh Mehta, 2nd Edition, Universities Press (India) Pvt. Ltd.

Reference Books:

1. An Introduction to Data Structures with Application, Jean-Paul Tremblay, Paul Sorenson, Second Edition.
2. Fundamentals of Data Structures and algorithms by C V Sastry, Rakesh Nayak, Ch. Raja Ramesh, IK Publications, new Delhi.
3. Data structures using C and C++, Langsam, Augenstein and Tanenbaum, PHI.
4. Problem solving with C++, The OOP, Fourth edition, W. Savitch, Pearson education.

Semester	III Sem	L	T	P	C	COURSE CODE
Regulation	V20	0	0	3	1.5	V20CSL05
Name of the Course	Linux Shell Scripting Lab					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate the basic knowledge of Linux commands and utilities by using Linux shell environment.(K3)
CO2: Experiment with the Concept of shell Programming on Files and Directories (K3)
CO3: Experiment with the Concept of shell Programming on File Permissions (K3)
CO4: Experiment with the Concept of shell Programming on Conditional Statements (K3)
CO5: Experiment with the Concept of shell Programming on Looping Statements (K3)

LIST OF EXPERIMENTS

1. Experiment the following Unix Commands:
 - a) **General Purpose Utilities:** cal, date,man,who.
 - b) **Directory Handling Commands:** pwd,cd,mkdir,rmdir.
 - c) **File Handling Utilities:** cat,cp,ls,rm,nl,wc
 - d) **Displaying Commands:** head, tail
 - e) **Filters:** cmp,comm.,diff,sort,uniq
 - f) **Disk Utilities:** du,df
2. Develop a Shell Program to Display all the words which are entered as command line arguments.
3. Develop a shell script that Changes Permissions of files in PWD as rwx for users.
4. Develop a shell script to print the list of all sub directories in the current directory.
5. Develop a Shell Program which receives any year from the keyboard and determine whether the year is leap year or not. If no argument is supplied the current year should be assumed.
6. Develop a shell script which takes two file names as arguments-If their contents are same then delete the second file.
7. Develop a shell script to print the given number in the reversed order.
8. Develop a shell script to print first 25 Fibonacci numbers.
9. Develop a shell script to print the Prime numbers between the specified range.
10. Develop a shell script to delete all lines containing the word 'unix' in the files supplied as arguments.
11. Develop a shell script Menu driven program which has the following options.
 - i) contents of /etc/passwd
 - ii) list of users who have currently logged in.
 - iii) present working directory.
 - iv) exit.

Text Books:

1. UNIX and Shell Programming: A Textbook, Behrouz A. Forouzan | Richard F. Gilberg, Cengage Learning.
2. UNIX: Concepts and Applications, Sumithaba Das, 4th Edition, Tata McGrawHill.
3. Unix & Shell Programming, M.G.Venkatesh Murthy, Pearson Education.
4. UNIX shells by example, 4th Edition Ellie Quigley, Pearson Education.

Semester	III Sem	L	T	P	C	COURSE CODE
Regulation	V20	0	3	0	MNC	V20ENT02
Name of the Course	Professional Communication Skills - I					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: At the end of the Course student will be able to:

- CO1:** Use vocabulary in regular chores of life with accuracy, make meaningful sentences, and describe people and their traits vividly. **(K3)**
- CO2:** Distinguish between places of pilgrimage and holiday spots; describe incidents, things and process; and frame questions, statements and expressions. **(K4)**
- CO3:** Demonstrate their knowledge of idioms which are similar to those of native speakers while speaking and writing and use phrases clearly and precisely to articulate their views that compare and contrast indianisms with native expressions and avoid common errors. **(K3)**
- CO4:** Employ the vocabulary of netizens with ease and walk through the letters and emails for effective official correspondence and infer the accurate meaning of the homophones that are often confusing. **(K3)**
- CO5:** Summarize their profile; introduce themselves as well as others by incorporating their accomplishments and Sketch stories and anecdotes in an interesting and engaging manner that arouses curiosity of the audience. **(K5)**

UNIT – I: BUILDING VOCABULARY FOR DAILY ACTIVITIES

NAMES: Things- Kitchen Utensils – Occupation- tools – spices- vegetables –flowers - sciences of study – Professions. Framing Questions – statements – expressions related to the Vocabulary taught.

PEOPLE: Describing people -Physical characteristics,-Mental attributes – various professions Framing Questions – statements – expressions related to the Vocabulary taught.

ACTIVITY: Related to the topics learnt in Unit – 1

UNIT – II: BUILDING VOCABULARY FOR PLACES, THINGS & PROCESS

PLACES: Describing favorite place – famous place- Places of Pilgrimage.

THINGS: Describing a thing- Describe an incident or an event.

PROCESS: Describe a process –Recipe – experiment –Entrance test application.

Framing Questions – statements – expressions related to the Vocabulary taught.

ACTIVITY: Related to the topics learnt in Unit – II.

UNIT – III:

NATIVE EXPRESSIONS – Idioms and Phrases – in day to day activities for different occasions - Usage written & spoken –

PHRASES with as—as expressions – used to compare & contrast

COMMON MISTAKES- in spoken & written

INDIANISMS- Most often used expressions – accepted in India – found place in Dictionary

ACTIVITY: Related to the topics learnt in Unit – III

UNIT –IV:

NET VOCABULARY: Acronyms and abbreviations that are most often used

HOMOPHONES : Words often confused – Spelling & Pronunciation

Letter Writing : Formal& Informal- Letters for all occasions

Email Writing : Business mails – project status mails – informative mails

ACTIVITY : Related to the topics learnt in Unit – IV

UNIT -V

SELF-INTRODUCTION: Basic information - Academic and personal - interests– strengths and weaknesses – goal.

PROFILE BUILDING: Resume writing – CV Building – Types

STORYTELLING WITH CREATIVITY: Reading and Narrating a story – narrating anecdotes

ACTIVITY : Related to the topics learnt in Unit – V

Reference Books:

- Lewis Norman, Word Power Made Easy (2008). Goyal Publishers & Distributors Pvt. Ltd.
- Sunita Mishra & C. Muralikrishna, Communication Skills for Engineers (2006). Dorling Kindersley (India) Pvt. Ltd., licensees of Pearson Education in South Asia.
- Chaturvedi PD & Chaturvedi Mukesh, Business Communication (2006). Dorling Kindersley (India) Pvt. Ltd., licensees of Pearson Education in South Asia.
- Joshi Manik, Popular English Idioms and Phrases: English Idiomatic Expressions (2013).
- Joshi Manik, Homonyms, Homophones and Homographs: Vocabulary Building (2014).
- Gupta S.C. A Handbook for Letter Writing (2018). Arihant Publishers
- Lisa McGrimmon, The Resume Writing Guide: A Step-by-Step Workbook for Creating a Winning Resume (2013). CareerChoiceGuide; 2nd edition.
- Sawhney, Clifford. Improve your Word Power (2013). V&S Publishers

Web References: (NET Vocabulary)

- <https://www.grammarly.com/blog/texting-abbreviations/>
- <https://www.slicktext.com/blog/2019/02/text-abbreviations-guide/>
- <https://www.webopedia.com/reference/text-abbreviations/>

Semester	IV Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CST06
Name of the Course	Design and Analysis of Algorithms					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: At the end of the Course student will be able to:

- CO1:** Demonstrate asymptotic notation and divide and conquer technique. **(K3)**
CO2: Use greedy technique to solve various problems. **(K3)**
CO3: Demonstrate dynamic programming technique to various problems. **(K3)**
CO4: Develop algorithms using backtracking technique. **(K3)**
CO5: Demonstrate branch and bound technique to various problems. **(K3)**

UNIT-I: Introduction: What is an Algorithm, Algorithm Specification-Pseudo code Conventions Recursive Algorithms, Performance Analysis-Space Complexity, Time Complexity, Asymptotic Notation, Practical Complexities, Performance Measurement.

Divide and Conquer: General Method, Binary Search, Finding the Maximum and Minimum, Merge Sort, Quick Sort-Performance Measurement,

UNIT-II: The Greedy Method: The General Method, Knapsack Problem, Job Sequencing with Deadlines, Minimum-cost Spanning Trees-Prim's Algorithm, Kruskal's Algorithms, Optimal Merge Patterns, Single Source Shortest Paths.

UNIT-III: Dynamic Programming: All Pairs Shortest Paths, Single Source Shortest paths General Weights, Explain Optimal Binary Search Trees, String Edition, 0/1 Knapsack, Reliability Design.

UNIT-IV: Backtracking: The General Method, 8-Queens Problem, Sum of Subsets, Graph Coloring, and Hamiltonian Cycles.

UNIT-V: Branch and Bound: The Method-Least cost (LC) Search, The 15-Puzzle: an Example, Control Abstraction for LC-Search, Bounding, FIFO Branch-and-Bound, LC Branch and Bound, 0/1 Knapsack Problem-LC Branch-and Bound Solution, FIFO Branch-and-Bound Solution, Traveling Salesperson. Basic Concepts of NP-hard and NP-complete problems.

Text Books:

1. Fundamentals of computer algorithms E. Horowitz S. Sahni, University Press.

Reference Books:

1. Introduction to Algorithms Thomas H. Cormen, PHI Learning.
2. The Design and Analysis of Computer Algorithms, Alfred V. Aho, John E. Hopcroft, Jeffrey D.Ullman.
3. Fundamentals of Data Structures and algorithms by C V Sastry, Rakesh Nayak, Ch. Raja Ramesh, Distributed by WILEY publications, New Delhi.
4. Algorithm Design, Jon Kleinberg, Pearson.

Semester	IV Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CST07
Name of the Course	Software Engineering					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate the Software Development life cycle Models. **(K3)**
CO2: Illustrate the Requirements engineering process and SRS document. **(K3)**
CO3: Develop the Software Architecture and Design Modeling. **(K3)**
CO4: Apply the Coding & Testing techniques and Risk management strategies. **(K3)**
CO5: Describe Project estimation techniques and Quality Management& Metrics. **(K2)**

UNIT-I: Software and Software Engineering: The Nature of Software, Software Engineering, Software Process, Software Engineering Practice, Software Myths. **Software process models:** Waterfall model, Prototyping, Iterative development, Unified process, RAD model, Spiral model, and agile process.

UNIT-II: Software Requirements: Functional and non-functional requirements, User requirements, System requirements, Interface specification, SRS document. **Requirements engineering process:** Feasibility studies, Requirements elicitation and analysis, Requirements validation, Requirements management.

UNIT-III: Software Architecture: Role of software architecture, Architecture views, components and connector view, Cohesion and Coupling, documenting architecture design. **Design:** Design concepts, Function-oriented design, object-oriented design, UML diagrams, and Data flow diagram.

UNIT-IV: Coding and Testing: Programming principles and guidelines, incrementally developing code. Testing concepts, testing process, Black-box & White-box testing. **Risk management:** Reactive vs. Proactive Risk strategies, Software risks, Risk identification, Risk projection, Risk refinement, RMMM Plan.

UNIT-V: Software Project Estimation& Maintenance: Decomposition techniques, Empirical Estimation Models, Maintenance Process, Reengineering, Configuration Management. Metrics for Products & Quality Management: Software Measurement, Metrics for software quality, Quality concepts, Software Reviews, Formal technical reviews, SEI-CMM Model, Six Sigma and ISO 9000 quality standards.

Text Books:

1. Software Engineering, A practitioner's Approach- Roger S.Pressman, 7th Edition, McGrawHill International Edition.
2. Software Engineering- Ian Sommerville, 9th Edition, Pearson education. Software Engineering, A Precise approach, PankajJalote, Wiley

Reference Books:

1. CMMI and Six Sigma: Partners in Process Improvement, Jeannine M. Sivi, M. Lynn Penn, Robert W. Stoddard, 1st edition, Addison Wesley;
2. Software Engineering principles and practice, WSJawadekar, 3rd Edition, TMH.

Semester	IV Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CST08
Name of the Course	Database Management Systems					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Describe Database systems, various Data models and Database architecture. (K2)
- CO2:** Develop various real time applications using Relational algebra and Relational calculus. (K3)
- CO3:** Apply various Normalization techniques to refine schema. (K3)
- CO4:** Explain Transaction management and Concurrency control. (K2)
- CO5:** Illustrate various Database indexing techniques. (K2)

UNIT-I: An Overview of Database Systems: Managing data, File systems verses DBMS, Advantages of DBMS, Data models, Levels of abstraction in a DBMS, Data independence, Structure of a DBMS, Client/Server Architecture, E.F.Codd Rules.

Database Design: Database design and ER Diagrams, Entities, Attributes, Entity sets, Relationships and Relationship sets, Conceptual design with ER Models.

UNIT-II: Relational Model: Integrity constraints over relations, Key constraints, Foreign key constraints, General constraints, Enforcing integrity constraints, Querying relational data

Relational Algebra: Selection and Projection, set operation, renaming, Joins, Division, Introduction to Views, destroying/altering Tables and Views.

Relational Calculus: Tuple Relational Calculus, Domain Relational Calculus.

UNIT-III: SQL Queries, Constraints and Triggers: The Form of Basic SQL Query, Union, Intersect, Except, Nested Queries, Aggregate Operators, Null Values, Complex Integrity Constraints in SQL, Triggers and active data bases.

Schema Refinement (Normalization): Problems caused by redundancy, Decompositions, purpose of Normalization, Schema refinement, Concept of functional dependency, Normal forms based on functional dependency (1NF, 2NF and 3NF), Concept of Surrogate key, Boyce-Codd Normal Form (BCNF), Lossless Join and Dependency preserving decomposition, Fourth Normal Form(4NF).

UNIT-IV: Transaction Management: Transaction, Properties of Transactions, Transaction Log, and Transaction management with SQL commit, rollback and savepoint.

Concurrency Control: Concurrency Control for Lost updates, Uncommitted data, Inconsistent retrievals and the Scheduler.

Concurrency Control with Locking Methods : Lock granularity, Lock types, Two phase locking for ensuring serializability, Deadlocks, Concurrency control with Time stamp ordering, Transaction recovery.

UNIT-V: Storage and Indexing: Overview of Storages and Indexing, Data on external storage, File organization and indexing, Clustered indexing, Primary and secondary indexes, Index data structures, Hash based indexing, Tree based indexing, Comparison of file organization

Text Books:

- Database Management Systems, Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition TATA McGraw Hill.
- An Introduction to Database Systems, C.J Date,A.Kannan,S.JSwamynathan 8th Edition, Pearson Education

Reference Books:

- Database Systems-Design, Implementation and Management, Peter Rob &Carlos Coronel 7th Edition, Course Technology Inc.
- Fundamentals of Database Systems, RamezElmasri,Shamkant B. Navathe ,7th Edition, Pearson Education.
- Database Systems - The Complete Book, Hector Garcia- Molina, Jeffry D Ullman, Jennifer Widom, 2nd Edition, Pearson.

Semester	IV Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20CST09
Name of the Course	Java Programming					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Describe Java Virtual Machine and Type casting. **(K2)**
CO2: Demonstrate Concepts like Constructors, Arrays, Nested Classes and Command Line Arguments. **(K3)**
CO3: Implement Concepts of Inheritance and Exception Handling **(K3)**
CO4: Develop programs on Multi-Threading and Files **(K3)**
CO5: Implement Event Handling and Swings. **(K3)**

UNIT-I: Introduction to Java: Introduction to Object Oriented Paradigm, Concepts of OOP, Applications of OOP, History of Java, Java Features, JVM, Program Structure. Variables, Primitive Data Types, Constants, String class, Primitive type conversion and Casting, Control Structures.

UNIT-II: Classes and Objects: Classes and objects, Class declaration, Creating objects, Methods, Constructors and Constructor Overloading, Importance of Static Keyword and Examples, this Keyword, Arrays, Command Line Arguments, Nested Classes, Garbage Collector.

UNIT-III: Inheritance and Exception Handling: Inheritance, super Keyword, final Keyword, Method Overriding and Abstract Class. Interfaces, Creating Packages, Using Packages, Importance of Class path. Exception Handling, Importance of try, catch, throw, throws and finally Block.

UNIT-IV: Multithreading and Files: Introduction, Thread Lifecycle, Creation of Threads, Thread Priorities, Thread Synchronization, Communication between Threads. Reading Data from Files and Writing Data to Files, Random Access Files.

UNIT-V: Event Handling and Swings: Introduction to AWT and Applets. Swings: Introduction, Components, Button, Label, Checkbox, List Boxes, Menu and Scrollbar, Layout Managers.

Event Handling : Event Delegation Model, Sources of Events, Event Listeners, Adapter Classes.

Text Books:

1. Java Programming, E. Balagurusamy, 4th Edition, TMH.
2. The complete Reference Java, 8th Edition, Herbert Schildt, TMH.
3. Introduction to java programming, Y Daniel Liang, 7 Edition, Pearson.

Reference books:

1. Core Java: An Integrated Approach , R Nageswara Rao, 7th Edition, Dream Tech
2. Head First Java , Kathy Sierra and Bert Bates, 2nd Edition O'reilly

Semester	IV Sem	L	T	P	C	COURSE CODE
Regulation	V20	3	0	0	3	V20MAT04
Name of the Course	Probability and Statistics					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Find the Expectation of Random variables.
- CO2:** Apply probability distribution to real time problems.
- CO3:** Plot a best fit curve to an experimental data and find the correlation and regression
- CO4:** Create good estimators to various parameters
- CO5:** Apply the principles of Statistical Inference to practical problems

UNIT-I: Random Variables and Expectation: Random Variables: Discrete and continuous - Probability function – density and distribution function, Expectation of a Random Variable, Moments, Chebychev's Inequality (Without proof).

UNIT-II: Probability Distributions: Probability distributions: Binomial, Poisson and Normal - Evaluation of statistical parameters: Mean, Variance and their properties, Introduction to Exponential, Gamma and Weibull distributions

UNIT-III: Bivariate Distributions: Curve fitting by the method of Least squares- Fitting of straight line, parabola and exponential curves, Simple Correlation and Regression – Rank correlation.

UNIT-IV: Sampling Distribution and Estimation: Introduction –Sampling distribution of means with known and unknown standard deviation. **Estimation:** Criteria of a good estimator, point and interval estimators for means and proportions

UNIT-V: Tests of Hypothesis: Introduction-Type-I, Type-II Errors, Maximum Error, one-tail, two-tail tests, Test of significance: Large sample test for single proportion, difference of proportions, single mean, difference of means.
Test of significance: Small sample test for single mean, difference of means and test of ratio of variances (F-Test) - Chi-square test for goodness of fit and independence of attributes.

Text Books:

1. **B. V. Ramana**, A text Book of Engineering Mathematics, Tata McGraw Hill.
2. **Miller & Freund's**, Probability & Statistics for Engineers – Eighth Edition, Richard. A. Johnson

References Books:

1. **S. Ross**, “A First Course in Probability”, Pearson Education India, 2002.
2. **Dr.T.S.R.Murthy**, Probability and Statistics for Engineers, BS Publications.
3. **T. Veerarajan**, “Engineering Mathematics”, Tata McGraw-Hill, New Delhi, 2010.

Semester	IV Sem	L	T	P	C	COURSE CODE
Regulation	V20	0	0	3	1.5	V20CSL06
Name of the Course	Statistical Visualization using R Lab					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: At the end of the Course student will be able to:

CO1: Employ math and simulation in R. (K2)

CO2: Demonstrate various types of data structures in R. (K3)

CO3: Apply appropriate control structures to solve a particular Programming problem. (K3)

CO4: Use R to graphically visualize data and results of statistical calculations. (K3)

LIST OF EXPERIMENTS:

1. Demonstrate the basic math functions in R
2. Demonstrate Vector operations in R
3. Demonstrate Matrix operations in R
4. Demonstrate Array operations in R
5. Demonstrate Data frames in R
6. Demonstrate Lists in R
7. Illustrate the following controls statements in R
 - a. if and else
 - b. ifelse
 - c. switch
8. Demonstrate for and while loops in R
9. Demonstrate importing and exporting data using R
10. Illustrate the descriptive statistics using summary() in R
11. Demonstrate the following statistical distribution functions in R:
 - a. Normal Distribution
 - b. Binomial Distribution
 - c. Poisson Distribution
 - d. Chi Square Distribution
12. Illustrate the following basic graphics in R:
 - a. Bar plots
 - b. Pie Charts
 - c. Histograms
 - d. Kernel density plots
 - e. Boxplots
 - f. Dotplots
13. Illustrate the Correlation and Covariance analysis using R
14. Illustrate the different types of t-tests using R
15. Illustrate the ANOVA test using R

Text Books:

1. R for Everyone, Jared P Lander, Pearson
2. R in Action, Rob I Kabacoff, Manning

Reference Book:

1. The Art of R Programming, Norman Matloff, No Starch Press

Semester	IV Sem	L	T	P	C	COURSE CODE
Regulation	V20	0	0	3	1.5	V20CSL07
Name of the Course	Data Base Management System Lab					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- | | | |
|-------------|---|------|
| CO1: | Construct SQL queries to perform different database operations. | (K3) |
| CO2: | Experiment with various constraints and Database Indexing Techniques. | (K3) |
| CO3: | Construct PL/SQL Cursors and Exceptions | (K3) |
| CO4: | Develop PL/SQL Functions, Procedures and Packages | (K3) |
| CO5: | Apply basic operations on collections of Mongo DB database | (K3) |

LIST OF EXPERIMENTS

Part-A

- Construct SQL queries to facilitate acquaintance of Built-In Functions, String Functions, Numeric Functions, Date Functions and Conversion Functions.
- Construct SQL queries using Operators.
- Construct SQL queries to Retrieve and Change Data: Select, Insert, Delete and Update
- Construct SQL queries using GroupBy, OrderBy, and Having Clauses.
- Construct SQL queries on Controlling data: commit, rollback and savepoint
- Construct report using SQL*PLUS
- Construct SQL queries for Creating, Dropping and Altering Tables, Views and Constraints
- Construct SQL queries on Joins and Correlated Sub queries
- Demonstrate Index, Sequence and Synonym.
- Demonstrate Controlling access, locking rows for update and security features.

PL/SQL

- Demonstrate Basic Variables, Anchored Declarations, and Usage of Assignment Operation Using PL SQL block
- Demonstrate Bind and Substitution Variables using PL SQL block
- Demonstrate Control Structures in PL SQL
- Demonstrate Cursors, Exception and Composite Data Types in PL SQL.
- Demonstrate Procedures, Functions, and Packages in PLSQL.

Part-B

- Demonstrate the installation of Mongo DB database.
- Demonstrate Creating and dropping database, collection in MongoDB.
- Demonstrate Insertion, updation and deletion operations in MongoDB database.
- Construct queries for Projection, limiting records, sorting records and aggregation in MongoDB database.

Textbooks:

- Oracle Database 11g The Complete Reference by Oracle Press, Kevin Loney
- Database Systems Using Oracle, Nilesch Shah, 2nd Edition, PHI.
- Introduction to SQL, Rick F Van der Lans, 4th Edition, Pearson Education.

Reference Books:

- Oracle PL/SQL Interactive Workbook, B. Rosenzweig and E. Silvestrova, 2nd Edition, Pearson Education.
- SQL & PL/SQL for Oracle 10g, Black Book, Dr. P.S. Deshpande, DreamTech.

Semester	IV Sem	L	T	P	C	COURSE CODE
Regulation	V20	0	0	3	1.5	V20CSL08
Name of the Course	Java Programming Lab					
Branch	Common to CSE&CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate Programs on Classes, Objects, Constructors and Arrays. **(K3)**
CO2: Demonstrate Inheritance and Exception Handling. **(K3)**
CO3: Implement programs on Multi-Threading and File Handling. **(K3)**
CO4: Implement Event handling using Swings. **(K3)**

LIST OF EXPERIMENTS

1. Develop programs on Control Structures and Type Conversions in java.
2. Develop programs using various String handling functions
3. Construct programs using the following concepts:
a) Classes & Objects b) Usage of static c) Constructors
4. Construct programs using the following concepts.
a) Arrays b) Nested Classes c) Command Line Arguments
5. Construct programs using the following concepts.
a) Inheritance b) Usage of super c) Method Overriding
6. Construct programs using the following concepts.
a) Usage of final b) Abstract class c) Interfaces
7. Implement the programs using the concepts
a) Packages b) Exception Handling.
8. Implement the programs on Multi-Threading.
a) Multiple Threads on Single Object b) Thread Deadlock
9. Construct a program that shows Inter-thread Communication
10. Construct programs to perform read and write operations on files.
a) Sequential Files b) Random Access files
11. Develop GUI using Swings.
12. Construct programs on Event Handling using Listener Interfaces.

Text Books:

1. The complete Reference Java, 8th Edition, Herbert Schildt, TMH.
2. Introduction to java programming, Y Daniel Liang, 7 Edition, Pearson.

Semester	IV Sem	L	T	P	C	COURSE CODE
Regulation	V20	-	2+2	-	MNC	V20ENT03
Name of the Course	Professional Communication Skills - II					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1** Demonstrate grammatical competence, analyze noun and pronoun dispositions, classify various kinds of verbs, adjectives and adverbs and identify errors in sentences; distinguish the subtle meanings of various words in different contexts, recognize similar words as well as words with contrast meanings and use them appropriately. **(K3)**
- CO2** Organize individual words into one whole sentence using new vocabulary and focus on the error analysis of prepositions and conjunctions, build conversations which befit the situations and develop pre-reading strategies to improve comprehension skills. Distinguish and acquire knowledge of using words of the same category in a sentence and learn new words that promote communicative finesse. Find errors in sentences where the modifiers are misplaced and put them at the appropriate place, use hit pair words and send an email that is concise and lucid. **(K3)**
- CO3** Recognize the easiest and best possible way of solving problems in the area of Number and Letter Series, Analogy, Classification, Coding & Decoding Symbols, Ranking and Analytical Reasoning. **(K4)**
- CO4** Investigate the different types of logics involved in Mirror and Water Images, Logical Reasoning & Arithmetic Reasoning. **(K4)**
- CO5** Find the common traps in the questions and errors likely to be made from the concepts of Blood Relations, Directions, Average, Clock and Calendar, Data Sufficiency, Permutations-Combinations and Probability. **(K3)**

UNIT – I

ERROR ANALYSIS: Nouns & Pronouns – Singular & Plural – Kinds of Nouns & Pronouns- Collective Nouns - Personal and Reflexive Pronouns. Subject – Verb agreement. Adjectives – Adverbs – role of modifiers – place of Adjectives– Adverbs of frequency.

VOCABULARY : Word Power Made Easy Sessions 15- 30, Antonyms and Synonyms and One word substitutes

EXPANSION OF PROVERBS: Meaning – interpretation – explanation.

UNIT – II

ERROR ANALYSIS: Prepositions - kinds of prepositions –appropriate use - conjunctions –subordinating– coordinating.

ROLE PLAY: Day to day situations - practical approach – real life experiences.

READING COMPREHENSION: Reading as a skill – quick reading - analyzing – answering - Skimming – scanning - summarizing – problem solving.

ERROR ANALYSIS: Parallel grammatical forms – same grammatical structures. Dangling modifiers – misplacement of modifiers – arrangement.

SENTENCE IMPROVEMENT: Better choice – error-free sentences – effective – syntax.

EMAIL WRITING: Format – method of exchanging – technicalities.

UNIT – III

Number And Letter Series, Coding S.No. & Decoding, Analogy, Classification Ranking. (K1)

Problems of how to find the next number in the series, Finding the missing number and related sums, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Finding odd one out from groups, Identify the rank in different places.

UNIT-IV

Problems On Ages& Numbers, Mirror And Water Images, Logical Reasoning & Arithmetic Reasoning.(K4)

Definition and concept of Venn Diagram – its applications. statements – Affirmations, Denials and Contradictions. Sums related to Ages & numbers. Problems on ages with different logics. Identifying the images of water and Mirror.

UNIT-V

Blood Relations, Directions, Average, Clock And Calendar, Data Sufficiency, Permutations-Combinations And Probability.(K3)

Deriving the formula to find the angle between hands for the given time, History of calendar-, Finding the day for the given date, Problems related to directions. Difference between words Permutation and Combinations – Various cases - Real Time Scenarios. Concept of Probability – - Conjunctions – Rules & Cases of Probability.

References

1. Verma Shalini. Common Errors In English (2016). S Chand & Company
2. Sharon Weiner Green M.A & Ira K. Wolf Ph.D. Barron's GRE (2015). Barrons Educational Series
3. Paul D.S. Advanced English Grammar with Answers (2007) Published by Cambridge University Press..
4. Work book -1 on Aptitude Prepared by T & P cell, Sri Vasavi Engineering College.
5. Kundan & Tyra. Magical Book on Quicker Maths (20013). Published by Tyra & Kundan
6. Kundan & Tyra. Practice Book on Quicker Maths (2009). Published by Tyra & Kundan
7. R.S. Agarwal . Non Verbal Reasoning. Sultan Chand Publications

Web References

<https://www.indiabix.com/>

<https://www.campusgate.co.in/>

<https://www.questionpaper.org/>